GRE Quantitative Reasoning

모의고사 Set 6 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 6 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
답															

Section 1 12 Questions · 21 Minutes

Question 1 대소비교

353. Quantity A: The remainder when 7^{22} is divided by 5
Quantity B: 4

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

355. d is a integer greater than 1.
Quantity A: $(d^2)!$
Quantity B: $(d!)^2$

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

876. List M: r+3, r+2, r+1, r, r-1, r-2, r-3
Quantity A: The standard deviation of the numbers in list M
Quantity B: r

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

347. The probability that a person will succeed at a particular task is p.
Quantity A: p(1-p)
Quantity B: 0.4

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

1297. If p is an even integer, which of the following must be an odd integer?
A.3p/2
B.3p/2 + 1
C.$3p^2/2$
D.$3p^2/2 + 1$
E.p^3

A B C D E

Questions 6 to 8 are based on the following data.

ON-TIME ARRIVALS AND CONSUMER COMPLAINTS
FOR MAJOR UNITED STATES AIRLINES, OCTOBER 2002

Airline	Percent of On-Time Arrivals in October 2002	Consumer Complaints per Million Passengers	
		Number in October 2002	Percent Decrease Since October 2001
A	80.2%	3.8	50%
B	87.5%	9.0	37%
C	82.4%	7.7	75%
D	78.4%	9.5	51%
E	82.1%	8.9	45%
F	86.4%	10.9	39%
G	82.6%	3.0	17%
H	88.0%	13.2	25%
I	87.9%	6.6	57%

Question 6 객관식

1283. Which of the airlines listed had the median percent of on-time arrivals in October 2002?
A.Airline A
B.Airline C
C.Airline E
D.Airline F
E.Airline G

A B C D E

Question 7 객관식

1284. For how many of the airlines listed was the number of arrivals that were not on time less than 15 percent of the total number of arrivals in October 2002?
A.One
B.Two
C.Three
D.Four
E.Five

A B C D E

Question 8 객관식

1285. For Airline H, what was the number of consumer complaints per million passengers in October 2001?
A.9.9
B.12.5
C.14.2
D.16.5
E.17.6

A B C D E

Question 9 객관식

708. Angle A measures r degrees. The complement of angle A measures 90-r degrees, and the supplement of angle A measures 180-r degrees. If the measure of angle A is greater than the measure of its complement and less than one-half the measure of its supplement, which of the following gives all possible values of r?
A $0 < r < 45$
B $45 < r < 60$
C $60 < r < 90$
D $90 < r < 180$
E $180 < r < 270$

A B C D E

Question 10 주관식

386. Select 2 students from A, B, C group and each group can only select no more than 1 student. Group A has 10 students, Group B has 7 students, while Group C has 8 students. In how many ways can these 2 students be selected?

Question 11 복수선택

331. Which of the following could be the value of x to make sure that x^3-x is divisible by 10?
Indicate all such values.
A480
B481
C482
D483
E484
F485
G486
H487

Indicate all that apply. A B C D E F G H

Question 12 객관식

1061. R_1: 13.2, 14.2, 15.2, 16.2, 17.2
R_2: (13)(0.2), (14)(0.2), (15)(0.2), (16)(0.2), (17)(0.2)
R_3:13/0.2,14/0.2,15/0.2,16/0.2,17/0.2
If s_1, s_2, and s_3 are the standard deviations of lists R_1, R_2, and R_3 respectively, which of the following shows the standard deviations in order from least to greatest?

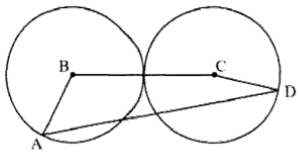
A s_1, s_2, s_3
B s_2, s_1, s_3
C s_2, s_3, s_1
D s_3, s_1, s_2
E s_3, s_2, s_1

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

1209.

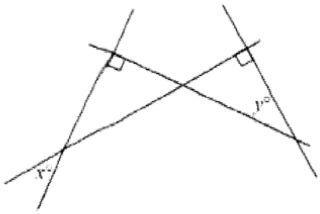


The two circles have centers at B and C, respectively, and are mutually tangent. Each circle has radius r.
Quantity A: The perimeter of quadrilateral ABCD
Quantity B: 8r

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

1227.



Quantity A: x
Quantity B: y

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

1163. Each of a, b, c, and d is a positive integer less than 10, and the product of a, b, c and d is 945.
Quantity A: The sum of a, b, c and d
Quantity B: 24

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1208. k is a positive integer
Quantity A: The number of prime numbers from k to k+10
Quantity B: The number of prime numbers from k+10 to k+20

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

334. In pentagon ABCDE, points M, N, P ,Q, and R are the midpoints of sides AB, BC, CD, DE, and EA, respectively.
Quantity A: The perimeter of pentagon ABCDE
Quantity B: The perimeter of pentagon MNPOR

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

513. A certain store sells circular rugs at a constant price per square foot. If a circular rug with diameter 5 feet costs $250, what is the cost of a circular rug with diameter 9 feet?
A$450
B$810
C$900
D$1620
E$1800

(A) (B) (C) (D) (E)

Question 7 객관식

527. The larger of two right circular cylindrical containers has an inside radius of 3 inches, and the smaller has an inside radius of 1.5 inches; each of the containers has an inside height of 10 inches. If the smaller container were filled with water until the depth of the water was 9 inches, and then its water were poured into the empty larger container, what would be the depth of the water, in inches, in the larger container?
A2.00
B2.25
C2.50
D2.75
E4.50

(A) (B) (C) (D) (E)

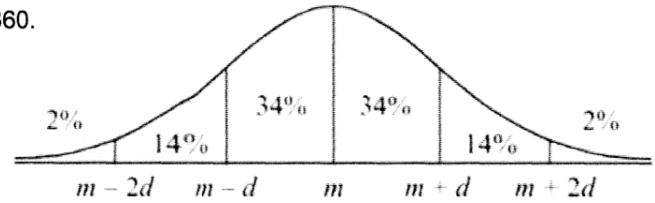
Question 8 객관식

1034. Set P consists of all the integers from 1 to 100, inclusive. Which of the following sets contains the greatest number of integers?
AThe integers of P that are multiples of 3
BThe integers of P that are multiples of 4
CThe integers of P that are multiples of 5
DThe integers of P that are multiples of both 3 and 4
EThe integers of P that are multiples of either 4 or 5

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 복수선택

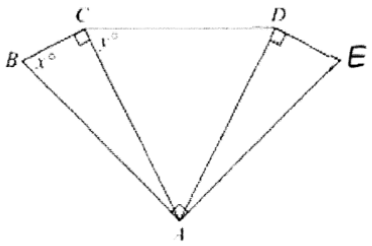
860.



The figure above shows a normal distribution with mean m and standard deviation d, including approximate percents of the distribution corresponding to the six regions shown.
The lengths of phone calls made on a certain weekend by students at High School H are approximately normally distributed with a mean of 30 minutes and a standard deviation of 10 minutes. Which of the following statements must be true? Indicate all such statements.
AThe range of the lengths of the phone calls is less than 60 minutes.
BThe lengths of half of the phone calls are each greater than 40 minutes.
CThe length of a 35-minute phone call is 0.5 standard deviation from the mean of the lengths of the phone calls.

Indicate all that apply. [A] [B] [C]

Question 10 주관식



404. Triangles ABC and AED are congruent, where BC=DE. If y=63, what is the value of x?

Question 11 객관식

1077. What is the sum of the integers between -90 and 95, inclusive?
A5
B185
C465
D4,275
E4,560

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 12 객관식

584. Each copy of a certain textbook in a college bookstore is classified as new or used. The bookstore sells each new copy for $90 and each used copy for another fixed price. For all the copies sold last week, the total revenue from the sales of new copies was equal to the total revenue from the sales of used copies, and the average (arithmetic mean) price per copy was $60. What is the price of each used copy?
A$30
B$40
C$45
D$72
E$75

(A) (B) (C) (D) (E)

Question 13 복수선택

497. The number of points of intersection between an equilateral triangle and a square could be (such that none of the sides would be overlapping)?
Indicate all that is/are true.
A1
B3
C5
D6
E7
F9

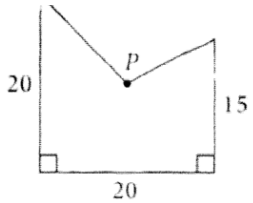
Indicate all that apply. [A] [B] [C] [D] [E] [F]

Question 14 객관식

603. A 100-dollar bill is exchanged at a bank for bills in denominations of 5, 10, and 20 dollars. If each of the three types of bills is received, what is the least number of bills that could be received?
AFive
BSix
CSeven
DEight
ENine

(A) (B) (C) (D) (E)

Question 15 객관식



465. The figure above represents the surface of a wall with an irregular shape, where all measurements are in meters and point P is 10 meters from the bottom edge and 10 meters from the left edge. The surface is to be painted, and one bucket of paint will cover 170 square meters of the surface. If the bucket of paint will cover the part of the surface from the left edge to a vertical line that is x meters from the left edge, which of the following is true?
A8 < x < 9
B9 < x < 10
C10 < x < 11
D11 < x < 12
E12 < x < 13

(A) (B) (C) (D) (E)